



PSOC™ 4 CAPSENSE™ using PSOC™ 4000T and PSOC™ 4100T Plus

Customer training workshop (CTW)

December 2025



Prerequisites

- Basic hands-on with ModusToolbox™
- Basic understanding of PSOC™ 4xxx device architecture
- Install or update the following on your PC:
 - ModusToolbox™ Tools Package v3.5.0
 - Python v3.12.x (Python v3.8.x to v3.12.x are supported by ModusToolbox™ v3.5.0)
 - ModusToolbox™ Eclipse IDE 3.5
 - ModusToolbox™ Programming Tools 1.4
 - ModusToolbox™ CAPSENSE™ and Multi-sense Pack

Agenda

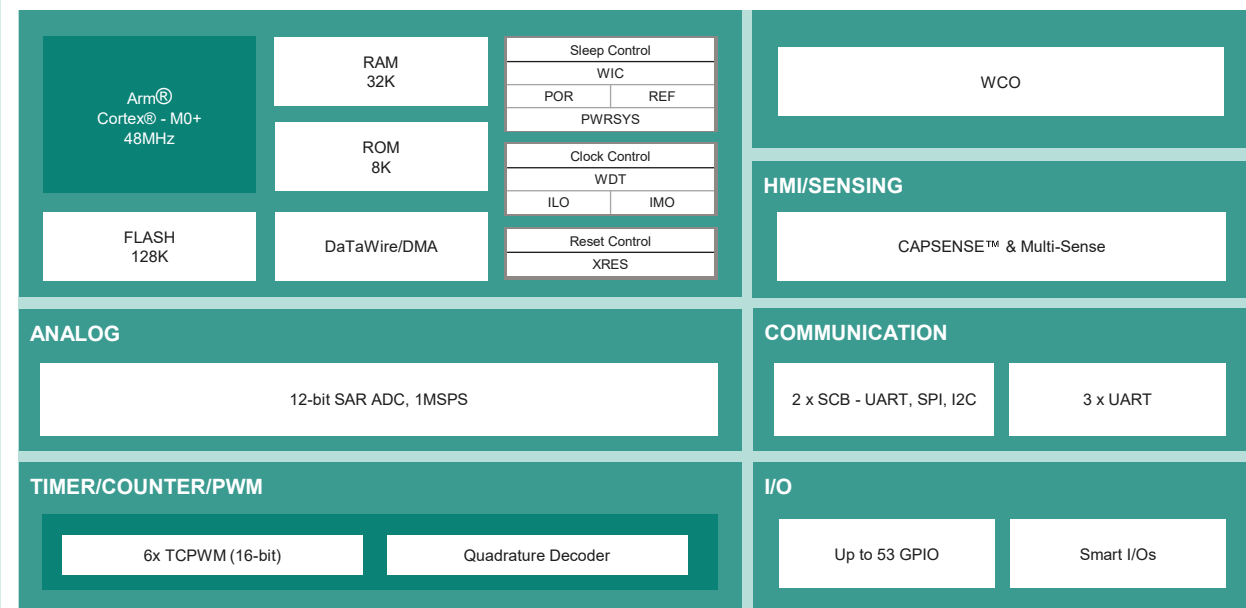
1	Introduction to the PSOC™ 4100T Plus	4
2	What's new in MSCLP	8
3	Low-power CAPSENSE™ design	14
4	Introduce the PSOC™ 4100T Plus proto kit	21
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32-bit MCU for multi-sense control

Block diagram – PSOC™ 4100T Plus



Applications

Home appliances, smart home, wearables, hearables, and other industrial and consumer applications.

Features

32-bit MCU subsystem

- 48 MHz Arm® Cortex®-M0+ CPU
- 128 KB flash and 32 KB SRAM
- DMA

Multi-sense with next-generation CAPSENSE™

- Fifth Gen CAPSENSE™ block with multi-sense, "Always-ON" sensing enabling ultra-low-power consumption
- Supports self-cap, mutual-cap and inductive sensing technologies
- Supports up to 32 sensors

Programmable digital blocks

- Six 16-bit timer/counter/pulse-width modulator (TCPWM) blocks
- Two serial communication blocks (SCBs) that are configurable as I²C, SPI, or UART
- Three dedicated UART interfaces

I/O subsystem

- Up to 53 GPIOs, including 32 sensors

Packages

- 48-QFN, 44-TQFP, 48-TQFP, and 64-TQFP

Other

- Main clock (IMO) 48 MHz with $\pm 1\%$ accuracy
- Enhanced ESD: 6.5 KV HBM, 2 KV CDM

CAPSENSE™ success with MSCLP 5th Gen

Parameter	CSDv2 (4 th Gen)	MSCv3 (5 th Gen) PSOC™ 4100S Max	MSCLP (5 th Gen with low-power)		Improvements
			PSOC™4000T	PSOC™4100T Plus	
Converter resolution (ENOB – Effective number of bits)	10-bit	14-bit	14-bit	14-bit	MSCv3, MSCLP 10x high SNR enables: – High performance sensors – Long distance proximity and SAR sensor – Interoperable and scalable low-cost touchscreen – High performance small screens mid-range performance large screens – Complex UI interfaces
External noise immunity	Medium	High	High	High	–
Autonomous scan (DMA, RAM)	N/A	Yes (RAM / DMA)	No DMA	Yes (RAM / DMA*)	~40% more CPU bandwidth for application processing (PSOC™ 4) ~20% faster HMI touch response rate
Multi-channel sensing	N/A	Yes	N/A	N/A	Enable large and complex HMIs (on connected devices)
Multi-chip sensing	N/A	Yes	N/A	N/A	Enable large and complex HMIs (on industrial large screens)
Wake-on-touch (WoT) / LP sensing	N/A	N/A	Yes	Yes	10x improvement of power consumption Enable low-power applications in wearable segment
Additional features		CIC2, HW IIR, CDAC Dither	All of MSCV3, Fine CDAC	All of MSCV3, Fine CDAC	–
CIC2	NA	Acc- 20-sample – 12-bit	Acc- 24-bit sample – 16-bit	Acc- 24-bit sample – 16-bit	–
External frame start	N/A	Yes	N/A	Yes	External Frame Start is add-on feature which helps to sync CAPSENSE™ scan with an external trigger.

* - With Application Layer Handling

Capacitive sensing technology- Comparison

Parameters	CSD V1 (3 rd Gen)	CSD V2 (4 th Gen)	MSCLP (5 th Gen)	Competition
Maximum supported CSD capacitance to achieve 5:1 SNR – hardware only	< 31.5pF	< 37.3pF	<230pF	28 pF for competition with best SNR
SNR (CSD)	5:1	6.5:1	48:1	–
Power supply rejection (DC)	-59 dB (GPIO)	-61 dB (GPIO)	-80 dB (GPIO)	N/A
Common mode rejection (AC)	N/A	-47 dB	-70 dB	N/A
Overlay thickness supported (CSD)	< 5 mm	5mm	18 mm	N/A
Active average current	–	156 μ A	117.8 μ A	–
ALR average current	–	40 μ A	29.2 μ A	–
Wake-on-touch (WoT) average current	–	21 μ A	3.8 μ A	–

Notes:

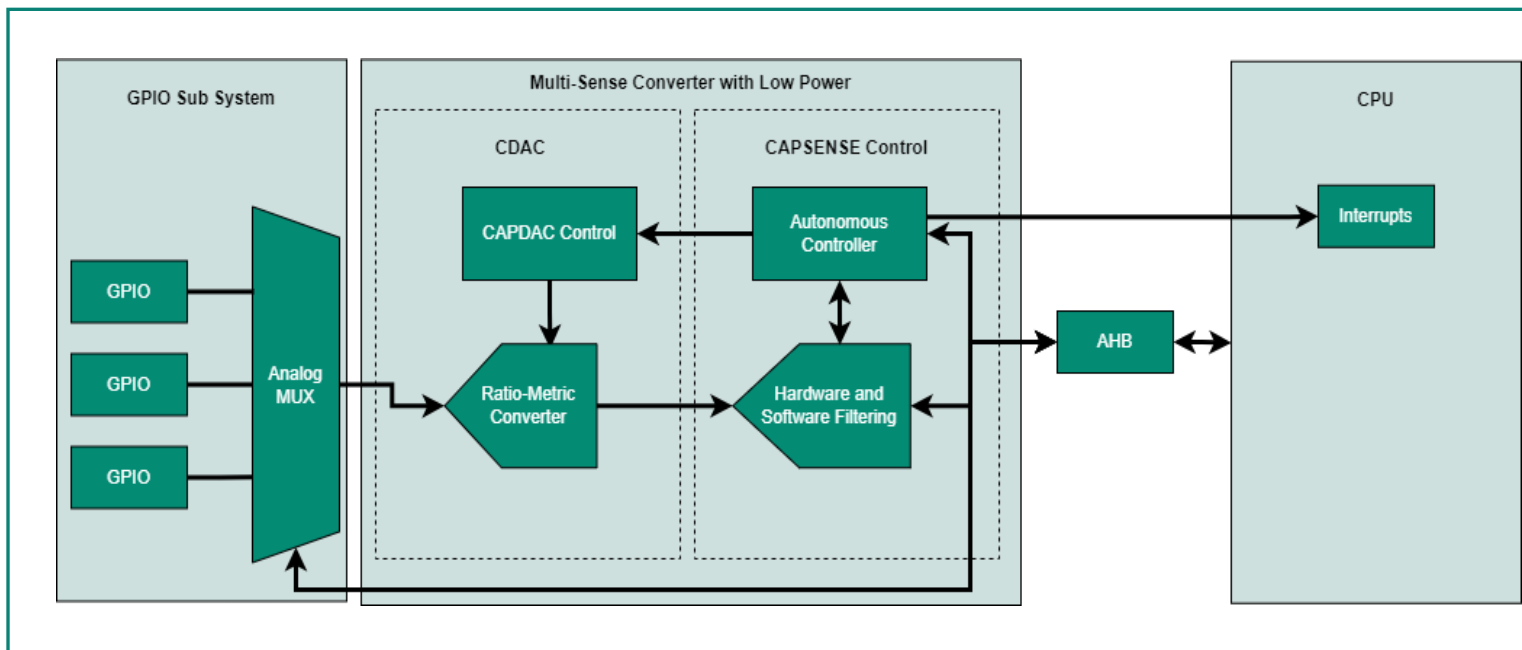
- Limited software filtering in above silicon results. Optimal software filtering significantly improves performance (up to 3x for CSD noise floor based on 7pF Si measurement).
- Key message: Optimal co-design of hardware and software including tweaks for custom application, based on several generations of experience.

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MSCLP 5th generation: Key updates

- Improved SNR
- Higher sensitivity
- Higher sensor capacitance range
- Easier layout
- Greater range of sensor sizes
- Wake-on-ouch (WoT)
- Ultra-low-power autonomous operation
- Improved liquid tolerance
- External sync

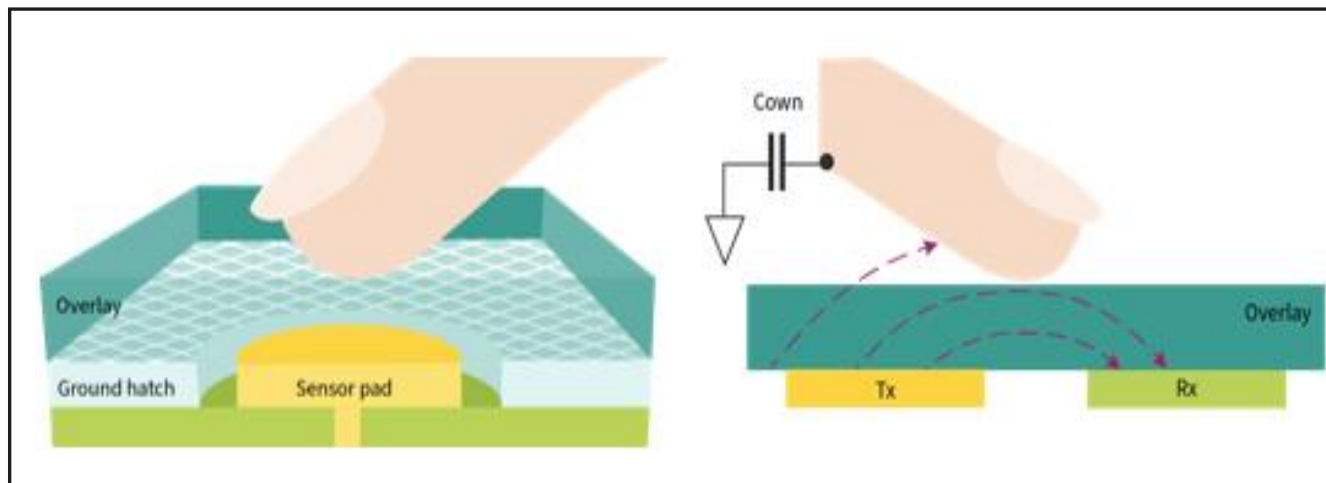


Improved SNR

- All new ratio-metric analog architecture
- Advanced hardware filtering
- Inherently lower internal noise sources
 - (System noise floor: 0.1 fF-rms with 8 pF input capacitor)
- **Benefits**
 - Shorter scan duration
 - Less software filtering
 - Lower power consumption
 - Robust operation in harsh environments
 - Ratio-metric design leading to improved DC PSRR

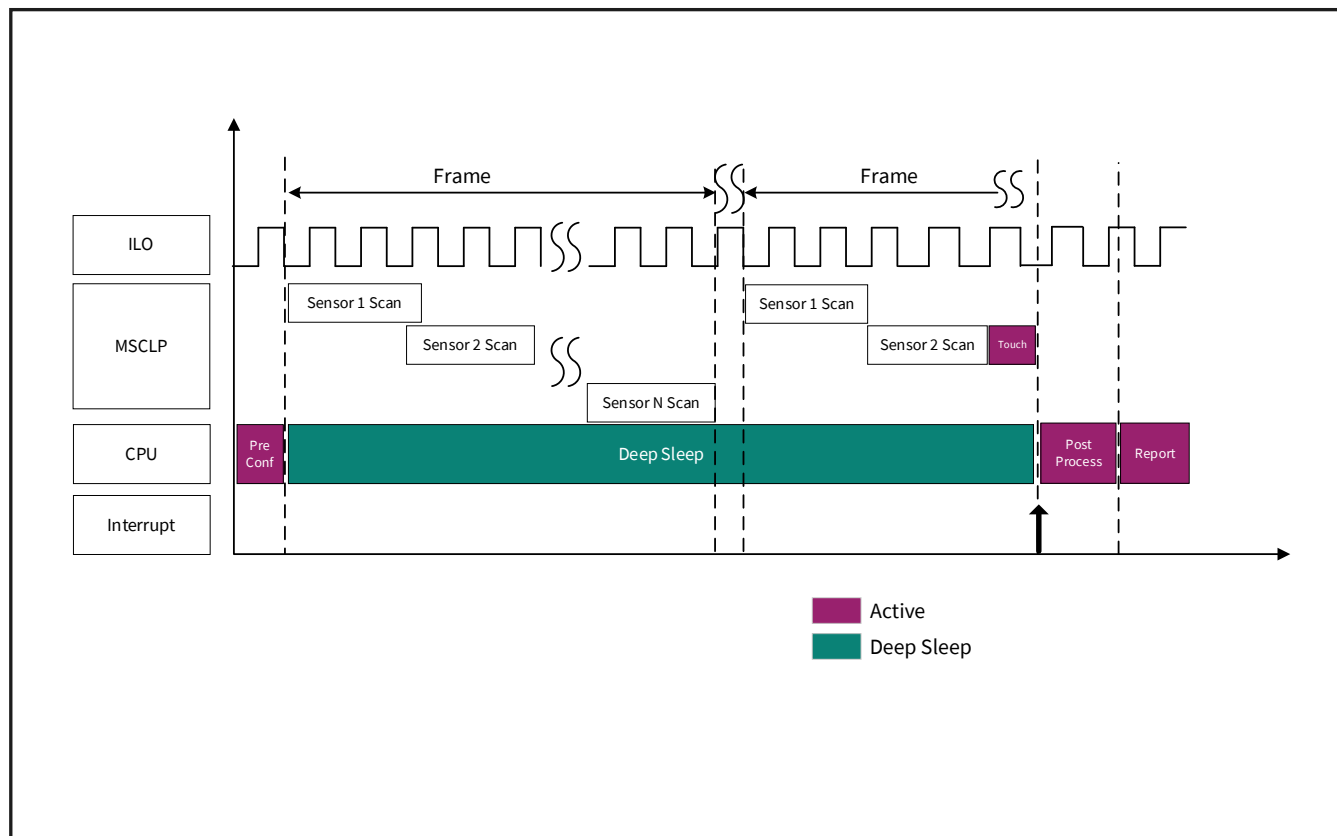
Higher sensitivity

- **Higher sensor capacitance range**
 - Input capacitance range for self-capacitance: 2 – 200 pF
 - Input capacitance range for mutual-capacitance: 0.5 – 30 pF
- **Benefits**
 - Support smaller and larger sensors
 - Higher proximity detection range
 - Much wider range of overlay thickness and materials



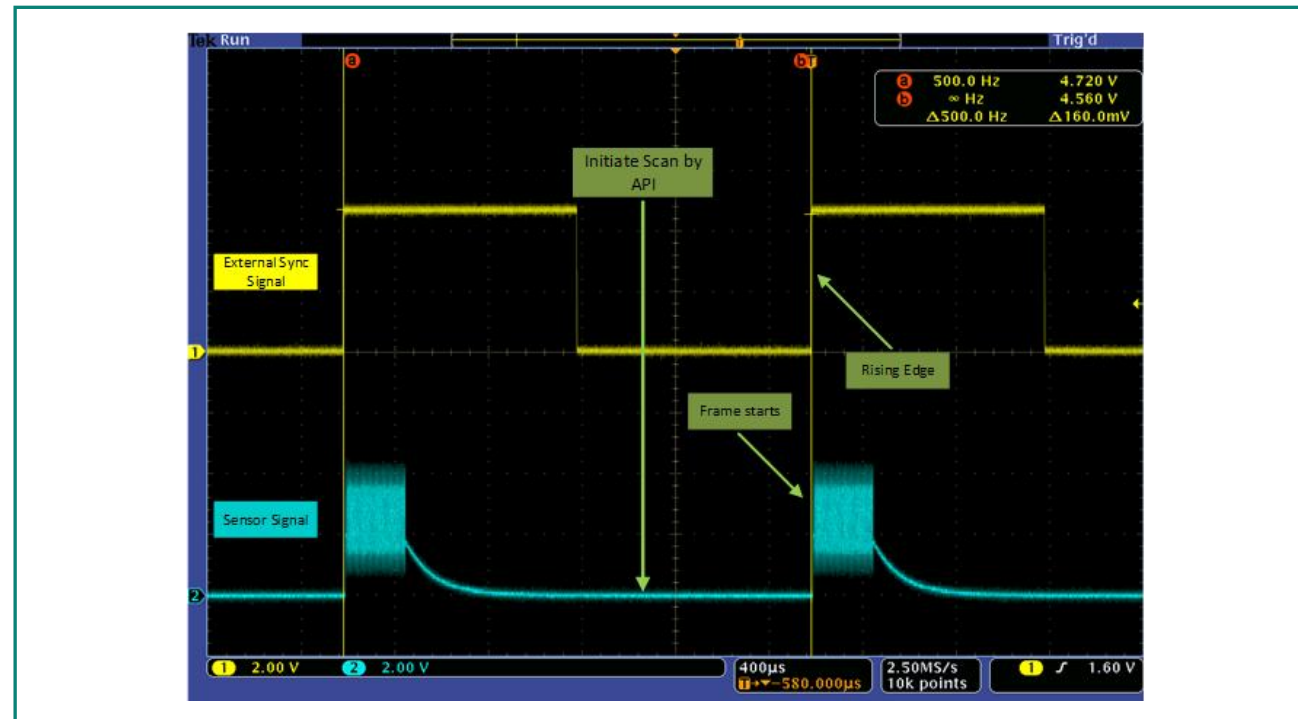
Wake-on-touch (WoT)

- Sensors scanned by HW at low refresh rate
- Results processed without CPU intervention
- Device can be kept in Deep Sleep (CPU, HFCLK OFF) while MSCLP is scanning
- Low-power interrupt generated by MSCLP hardware when touch is detected
- Benefits:
 - Device can be in Deep Sleep when there is no touch activity
 - Lower-power consumption



External frame start

- Synchronize start of frame scan to an external signal
- Initiated by a rising edge on ext_frm_start pin
- External signal can even be non-periodic
- Benefits
 - Sync with external noise source (for example, LCD)
 - Avoid the noise or keep the effect consistent from scan to scan
 - Scanning rate controlled by host

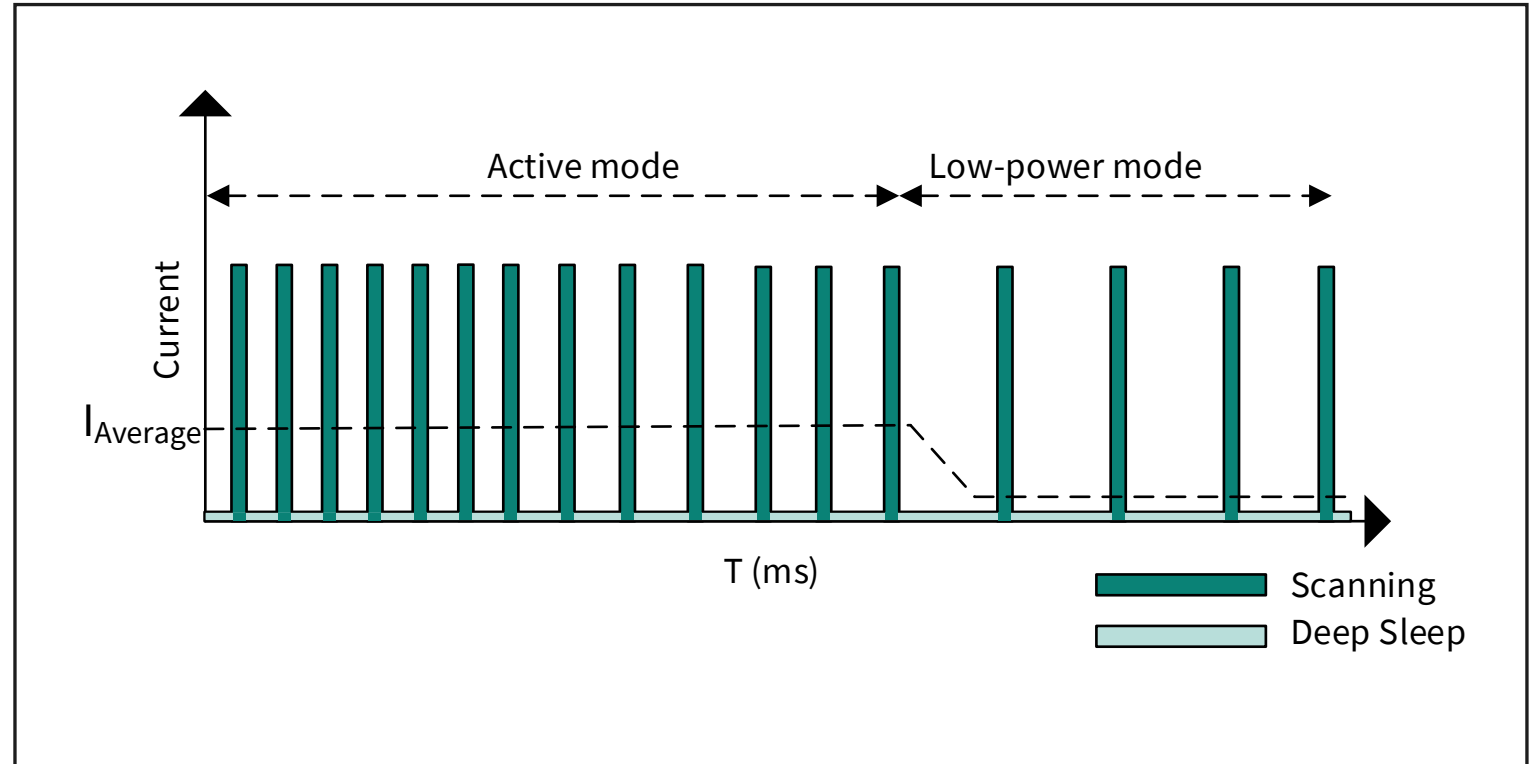


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Low-power CAPSENSE™ design

- Wearables are in low-power mode most of the time
- Current consumption in low-power mode significantly impacts battery life
- Touch detection is used to transition from low-power to Active state

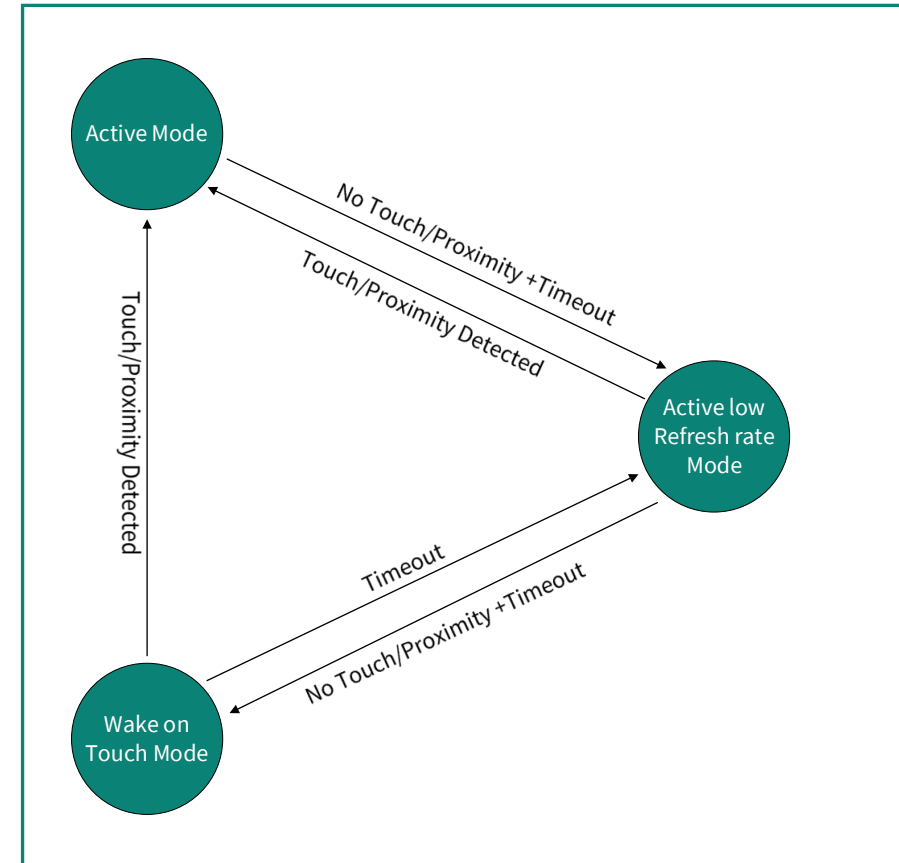


MSCLP – Low-power features

- MSCLP uses wake-on touch (WoT) mode to detect touch while the device is in Deep Sleep
- Autonomously scans several sensors without CPU intervention
 - The MSCLP has a dedicated 1024 bytes of SRAM
 - Active mode sensors
 - 21 sensors with HW IIR filter enabled
 - 36 sensors with HW IIR filter disabled
 - WoT mode sensors
 - 21 sensors HW IIR filter is always enabled for LP scanning
- Does not require the main IMO clock for scanning
 - Still requires CAPSENSE™ IMO
 - Main IMO is required for gestures like click, scroll, flick, swipe, and rotate
- Reduces power in the Active mode
- Enables 5-10x lower-power consumption in Low-power mode
- Enables 3-5x lower-power consumption in Active mode

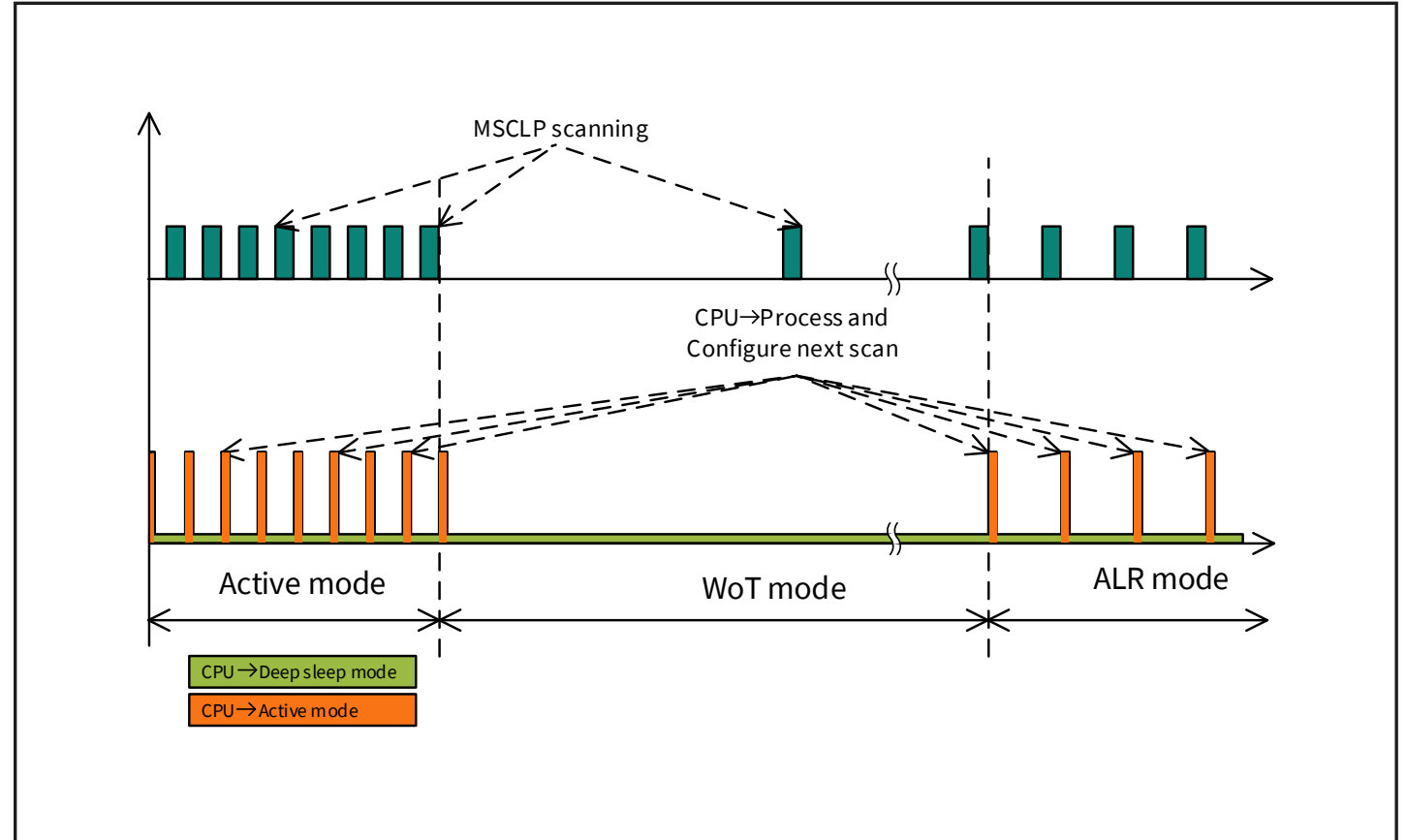
Application power modes

Application power modes	Scope	Refresh rate	Power consumption	Optimum device power mode
Active (max refresh rate)	All sensors	High	High	Active mode: To trigger scan and process Deep sleep mode : For frame scan and waiting (to achieve required refresh rate).
Active - low refresh rate (ALR)	All sensors	Low	Medium	Active mode: To trigger scan and process Deep Sleep mode: For frame scan and waiting (to achieve required refresh rate).
Wake on touch (WoT)	Few sensors or ganged sensors	Very Low	Low	Deep sleep mode: To scan, process, and wait.



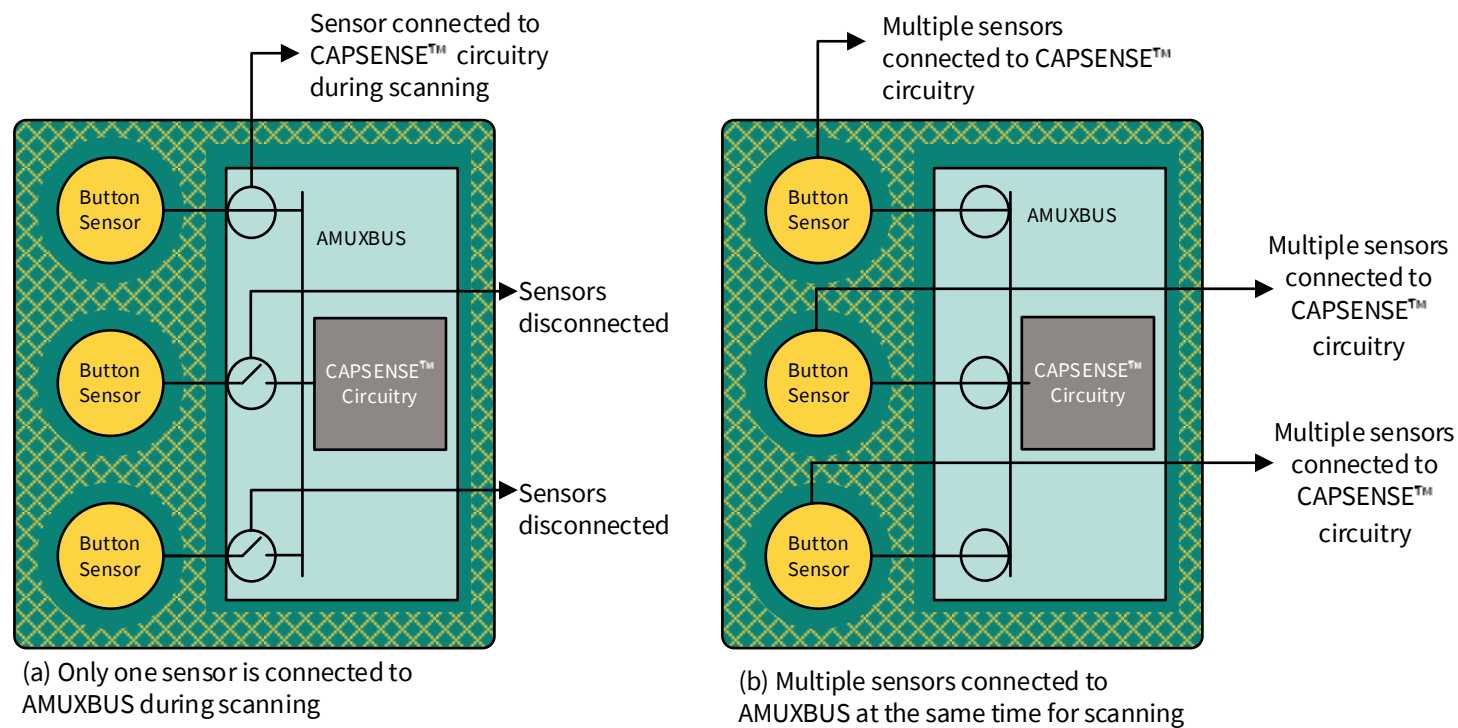
Wake-on-touch (WoT) sensing

- Autonomous scanning mode is the default scanning mode in fifth generation LP CAPSENSE™
- Even in Active application power modes, the device power mode can be put to deep sleep for optimal power consumption
- Sensors can be ganged in WoT mode to create a larger proximity sensor



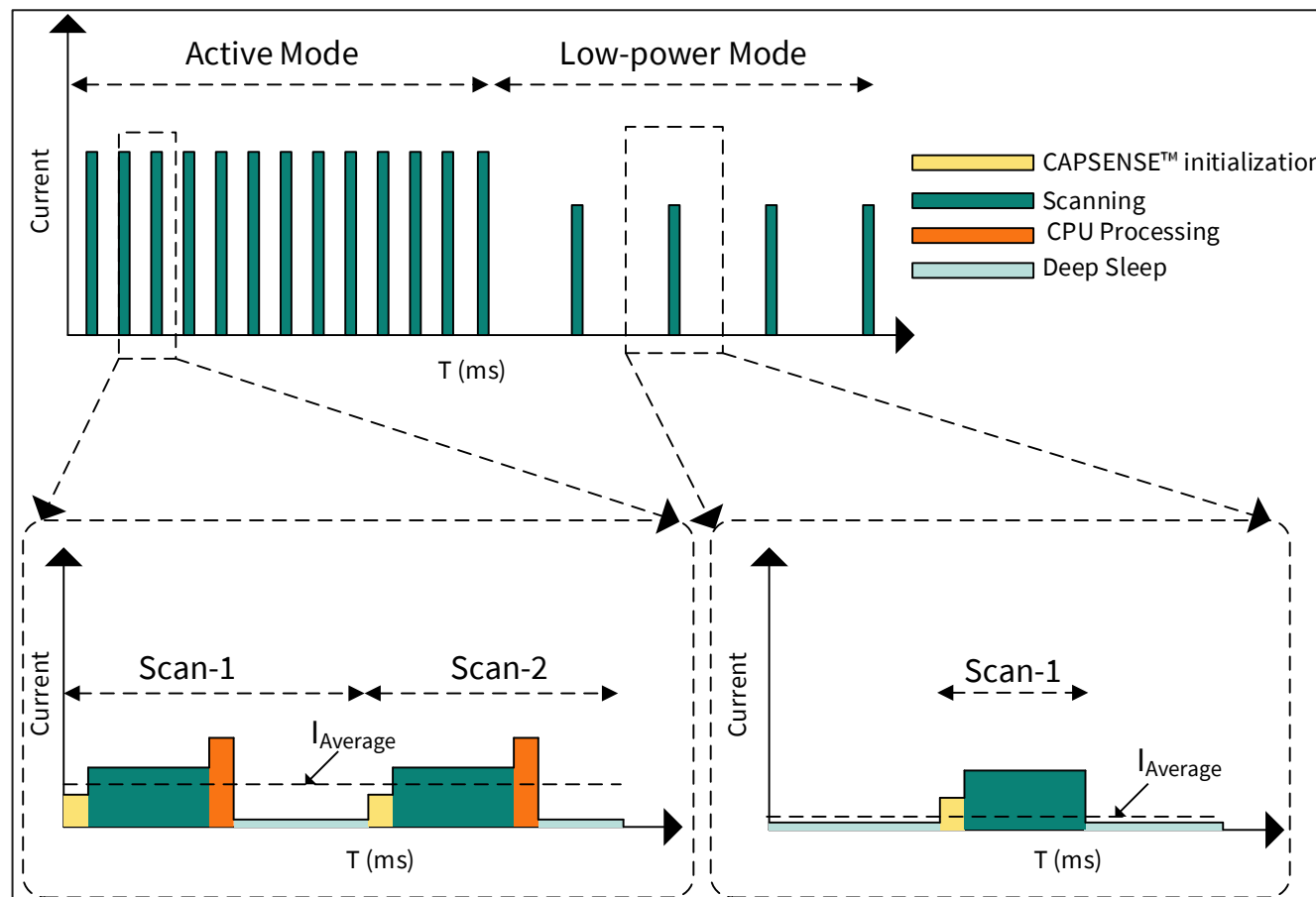
Sensor ganging

- Sensors can be ganged to create a larger proximity sensor
- This is especially beneficial for low-power sensors



Achieve low-power

- Scan duration
 - Depends on the C_p of the sensor
 - Max scan frequency limited by parasitic capacitance (C_p)
- Refresh rate and power modes
 - User experience and power
- Number of sensors in low-power mode
 - Low C_p reduces power
 - Dedicated wakeup button is ideal
 - Gang sensors helps to reduce power

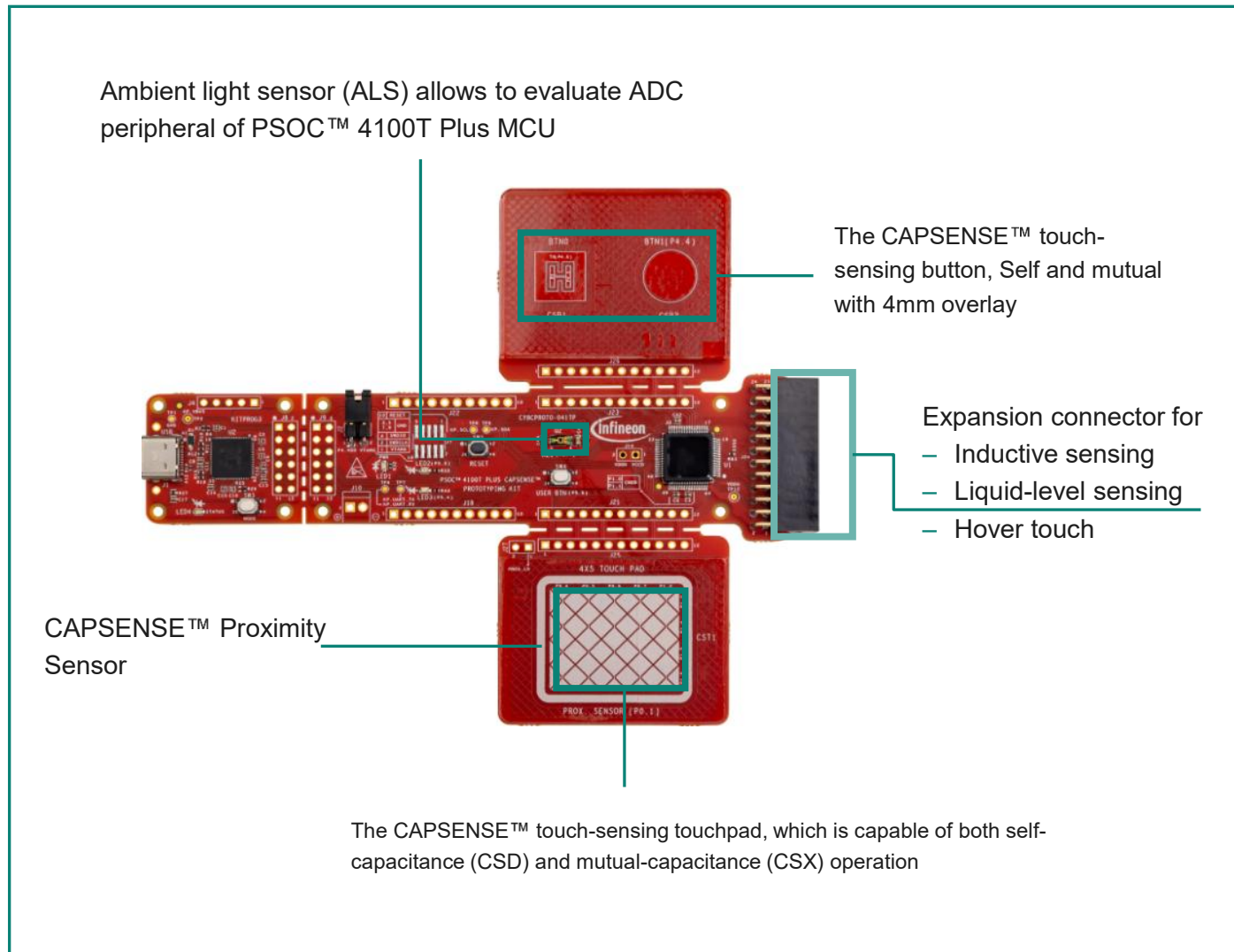


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CY8CPROTO-041TP - PSOC™ 4100T Plus Prototyping kit

Kit MPN	CY8CPROTO-041TP
Release schedule	Q2'2025 (Full Market Launch)
KitProg3	Yes
Autonomous operation without CPU	Yes
Ultra-low power Always-ON sensing	Yes
ADC measurements	Yes
CAPSENSE™ buttons	One self-cap button, one mutual-cap button
CAPSENSE™ trackpad	5 x 4 Touchpad, self and mutual sensing capable
CAPSENSE™ proximity	Yes
Inductive sensing	Through expansion board
Liquid-level sensing	Through expansion board
Expansion connectors	Yes, develop custom inductive and LLS boards or connect to expansion board of CY8CPROTO-040T-MS kit
Example code	CE240380, OOB example code
Best suited for	Traditional proto-typing kit that is easy to use, 32-bit MCU application development, connect external sensors on your own hardware and enables quick proof of concept development and evaluation of 5th Gen CAPSENSE™, Inductive Sensing and Liquid-level sensing



CY8CPROTO-041TP PSoC™ 4100T Plus CAPSENSE™ Prototyping Kit

– Low-power

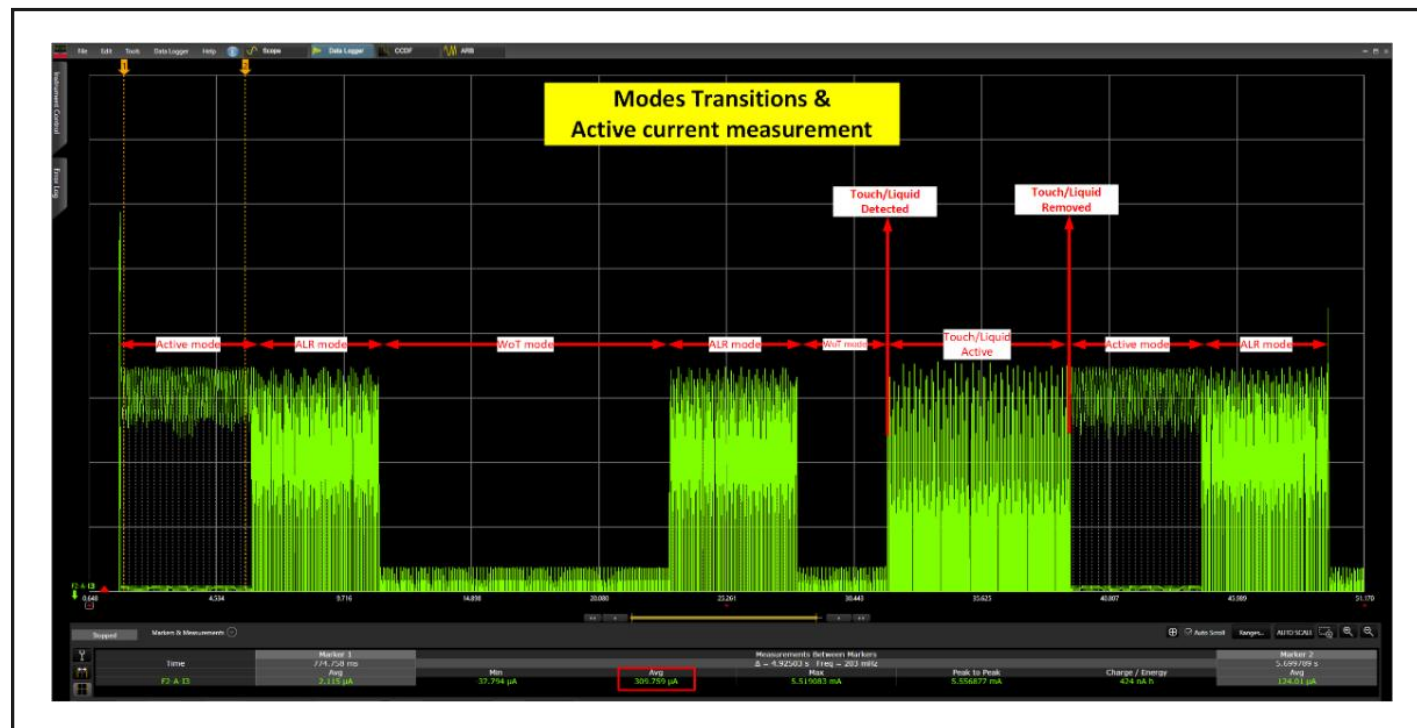
- After reset, the application transitions to low-power states if there is no user activity to reduce the power consumption
- If there is touch detection while in low-power states, the application transitions to active mode with highest refresh rate

Device current at different application states

Application State	Refresh rate (Hz)	Current (μA)*
WoT	16	5
Active Low Refresh Rate	32	26
Active	128	96

For more information on the specifics of the code example, see the [readme](#).

Power consumption profile of PSoC™ 4100T Plus



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CAPSENSE™ performance tuning

What will you learn ?

- How to set the initial hardware parameters
- How to set the sense clock frequency
- How to fine tune for the required SNR, power, and refresh rate
- How to tune the threshold parameters

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CAPSENSE™ low-power tuning

What will you learn ?

- How to use Deep Sleep in regular widget scanning
- How to optimize the refresh rate and power consumption
- How to use application power states to conserve power

Revision history

Document revision	Date	Description of change
**	2025-12-04	Initial release.



Thank you! Stay tuned for more trainings...

Join with us on our mission to drive decarbonization and digitalization Together.



